

ANSWER KEY - KEEP AWAY FROM QUIZ-TAKERS!

All Quizzes: Answers & Explanations

Each entry shows the correct letter and the explanation the app gives. Reading the explanation aloud after grading each question is half the fun.

Lesson 1 Quiz: The Isolette (pass mark 6/8)

1. **B** - Staying warm. A preemie has thin skin, almost no fat for insulation, can't shiver, and has very little brown fat — so without help, its temperature drops fast, and everything else gets worse from there.
2. **B** - The probe constantly reports skin temperature to the isolette, which warms or cools the air to keep the baby right on target. That automatic chasing-a-target trick is called servo control.
3. **B** - Newborns can't really shiver — their muscles are too small. Full-term babies use brown fat to make heat instead, but preemies are born before most of their brown fat has grown.
4. **A** - Brown fat burns fuel to make heat directly — a newborn's replacement for shivering. It grows mostly in the last weeks of pregnancy, so the earliest babies have almost none.
5. **C** - Evaporation — as water dries off the skin, it carries heat away, like when you're freezing right after getting out of a pool. That's why the tiniest preemies go straight into a plastic wrap before they even dry.
6. **B** - Extremely premature skin leaks water for days after birth, and evaporating water steals heat. Rainforest-level humidity inside the isolette stops the leak until the skin matures.
7. **C** - Kangaroo care is skin-to-skin holding. A parent's chest keeps the baby warm remarkably well, and it also steadies the heartbeat and breathing and helps with feeding. It's treatment that parents provide.
8. **B** - A cold baby spends precious sugar and oxygen making heat instead of growing. Blood sugar can crash, breathing struggles, and survival drops. That's why warmth is treated as seriously as medicine.

Lesson 2 Quiz: The Pulse Oximeter (pass mark 6/8)

1. **B** - Hemoglobin is the oxygen-carrying protein packed inside red blood cells — each one can grab four oxygen molecules, and each red cell holds about 270 million hemoglobins.
2. **C** - Hemoglobin absorbs red and infrared light differently depending on whether it's carrying oxygen. Comparing how much of each light makes it through the foot reveals the oxygen percentage — no poke needed.
3. **B** - Loaded hemoglobin is bright cherry red; empty hemoglobin is darker and bluish-red. That color difference is the whole secret the pulse oximeter exploits.
4. **C** - In healthy kids and adults, about 95-100% of hemoglobin is carrying oxygen. (It can never go above 100% — you can't fill a bus past full!)
5. **B** - Too much oxygen harms the delicate growing blood vessels in a preemie's eyes (ROP) and can injure the lungs. But too little is dangerous too — big studies found roughly 90-95% is the safest zone. Even good things have a right dose.
6. **B** - The trade-off was real: fewer eye problems, but more deaths. That's why most NICUs settled on roughly 90-95% — enough oxygen to live and grow, not so much that it poisons the eyes and lungs.
7. **A** - Look at the baby first! A pink, wiggly baby with a slipped probe is fine; a dusky, still baby needs help right now. Machines report; humans judge.

8. **C** - The oxygen-blindness epidemic — which affected baby Stevie Wonder, born six weeks early in 1950 — taught medicine to test every assumption. Even oxygen has a dose.

Lesson 3 Quiz: The Monitor (pass mark 6/8)

1. **A** - A patch of pacemaker cells at the top of the heart fires an electrical spark, and the heart muscle squeezes as the signal sweeps across it. Spark, squeeze, rest — about 100,000 times a day.

2. **B** - The heart's electrical sparks leak all the way out to the skin, and the three sticker leads detect them. The wiggly line they draw is the ECG — each spike is one heartbeat.

3. **C** - Newborns normally run about 100-160 beats per minute — roughly twice an adult's speed. Small hearts move a spoonful of blood per squeeze, so they squeeze often.

4. **A** - The brain's breathing control center is still under construction in preemies. The baby just... pauses. No choking, no struggle — which is exactly why a machine has to watch.

5. **B** - When breathing pauses (apnea), oxygen falls and the heart often slows down (bradycardia). These paired spells are so common that the unit just calls them A's and B's.

6. **B** - Touch usually does it — a gentle rub wakes the breathing center, the baby takes a big breath, and the numbers recover. Only stubborn spells need puffs of air from a bag and mask.

7. **C** - Carefully dosed caffeine keeps the immature breathing center alert. The famous CAP trial (~2,000 babies) showed it also protects lungs and helps development. Weird and wonderful!

8. **B** - A walnut-sized heart moves only a spoonful of blood per squeeze, so it squeezes often — and growing at rocket speed takes serious oxygen. As bodies grow, the numbers slow: newborn ~140, you ~90, adult ~70.

Lesson 4 Quiz: Blood Pressure (pass mark 6/8)

1. **B** - Blood pressure is the push of blood against the artery walls — like water pressure in a garden hose. That push is what drives deliveries of oxygen and fuel to every organ.

2. **C** - The top number is systolic: the pressure high point during the heart's squeeze. The bottom number (38) is diastolic: the low point while the heart refills. MAP is the average of the whole wave.

3. **B** - A tiny body needs less push to reach everywhere. Around 60-80/40-50 is healthy for a term newborn, versus about 120/80 for an adult. Adult pressure would be dangerously HIGH for a baby.

4. **B** - Born at 26 weeks doctors often worry below a MAP of about 26. But it's a rough rule, and good doctors also check pee output, warmth, and cap refill — they treat the baby, not just the number.

5. **A** - For a few days after birth, the umbilical cord stump still has open vessels. A soft catheter placed there gives a live pressure wave and painless blood samples — the cord becomes a lifeline again.

6. **B** - Urine output is one of the best simple clues that blood pressure and circulation are doing their real job: delivering blood to the organs. NICU teams measure every diaper — seriously, they weigh them.

7. **D** - Fluids, squeeze-medicines like dopamine, and cause-hunting are the real toolkit. (Cocoa's day comes later in life.) And if the baby is delivering blood well despite a lowish number, sometimes the smartest move is careful watching.

8. **B** - A preemie's brain is full of fragile, freshly built vessels. Crashes starve it; surges can rupture them and cause a brain bleed. Keeping pressure steady is quiet, hourly, brain-saving work.

Lesson 5 Quiz: CPAP & the Ventilator (pass mark 6/8)

1. **B** - Lungs are sponges made of hundreds of millions of tiny balloons called alveoli. Oxygen crosses their tissue-thin walls into the blood, and carbon dioxide crosses out.
2. **B** - Wet surfaces stick — surface tension constantly tries to collapse every alveolus. Surfactant's slippery coating cancels the stickiness so the balloons stay springy and open.
3. **B** - Babies make enough surfactant only around 34-36 weeks. Born earlier, their sticky alveoli collapse after every single breath, and re-opening them over and over is exhausting.
4. **B** - CPAP's steady air cushion means the alveoli never collapse all the way. The baby still breathes on its own, but every breath starts from 'partly inflated' — the easy part.
5. **A** - CPAP is a cushion; the ventilator is a pilot. When a baby can't do the breathing work even with CPAP, the ventilator takes over — as gently as possible, and weaned down as soon as possible.
6. **B** - Steroid shots given to the mother cross to the baby and speed up lung development — including surfactant production. Given in time, they raise survival before the baby's first breath.
7. **B** - In 1963 there was simply nothing that could fix sticky lungs — not for anyone. His death pushed RDS research forward, and by around 1990, surfactant treatment was saving babies just like him.
8. **A** - Mary Ellen Avery (with Jere Mead) found that babies who died of RDS lacked surfactant. Her 1959 discovery launched the thirty-year effort that turned RDS from the top killer of preemies into a treatable disease.

Lesson 6 Quiz: Feeding Tubes & IV Lines (pass mark 6/8)

1. **B** - Feeding is a brain skill: suck, swallow, breathe, in perfect rhythm, with the airway gate flipping correctly every time. Before ~32-34 weeks the wiring isn't there, so milk could end up in the lungs.
2. **B** - NG = naso-gastric, nose-to-stomach. Milk is pumped through it slowly and precisely, skipping the risky swallowing step — the baby digests normally. That's gavage feeding.
3. **B** - Total parenteral nutrition skips the gut entirely: a custom recipe recalculated every day and dripped into a vein. It's how the smallest, sickest babies grow before their gut is ready.
4. **B** - They're gut practice, not calories. Waking the intestines slowly and increasing feeds gradually is part of protecting against NEC.
5. **A** - Necrotizing enterocolitis is one of the most feared NICU diseases. Mild cases heal with gut rest and antibiotics; severe cases need emergency surgery, and some babies die of it. It's why every preemie belly is watched like a hawk.
6. **B** - Human milk brings germ-fighting antibodies and measurably reduces NEC compared with formula. When mom's milk isn't available, NICUs use tested, pasteurized donor milk from milk banks.
7. **B** - About 500 g — two sticks of butter, or half a big water bottle. NICU life is measured in grams, and every gram gained is celebrated.
8. **B** - Non-nutritive sucking lets the baby rehearse the sucking part of the suck-swallow-breathe stunt with zero risk, so the skill is readier when real bottle or breast practice starts around 32-34 weeks.

Lesson 7 Quiz: The Bili Lights (pass mark 6/8)

1. **B** - Red blood cells only live a few months. When the body breaks them down for recycling, the hemoglobin leaves behind a yellow chemical: bilirubin. (It's also the yellow stage of a healing bruise!)

2. **B** - A perfect storm: lots of bilirubin being made (spare red cells being retired) plus a liver that hasn't warmed up yet. Most newborns get at least a little yellow; preemies most of all.
3. **C** - Light energy literally bends the bilirubin molecule into water-friendly shapes like lumirubin, which wash out of the body without the slow liver's help. Photons as medicine.
4. **A** - The lights are spa-level bright, so soft shields protect the baby's eyes during the hours of treatment. The rest of the skin stays bare — more skin in the light means faster treatment.
5. **B** - At extreme levels bilirubin can cross into the brain and permanently damage movement and hearing centers. It's rare with good newborn care — which is exactly why every baby, everywhere, deserves a bilirubin check.
6. **B** - In 1956, Sister Jean Ward noticed sun-bathed preemies faded to normal except under blanket lines — and a blood sample left in sunlight lost bilirubin too. Noticing things is a doctor's superpower.
7. **A** - Pooping is literally how bilirubin exits, so frequent feeding is genuine treatment. And for dangerous extremes there's the rare emergency backup: exchange transfusion, swapping in donor blood.
8. **B** - Bilirubin normally rises and falls on a schedule over the first week. The same number can be fine or alarming depending on timing — so levels get plotted on an hour-by-hour chart.

Lesson 8 Quiz: Seeing Inside (pass mark 6/8)

1. **B** - An X-ray is a shadow picture: dense bone blocks the rays (white), soft organs dim them (gray), and air lets them sail through (black).
2. **B** - Ultrasound is echolocation: millions of sound pips too high to hear, echoes timed precisely, and a live moving picture painted from the bounces. No radiation at all.
3. **B** - The skull's plates haven't knitted together yet, leaving a bone-free diamond on top. Rest the wand there and the ultrasound sees the brain clearly. Nature left a door, and doctors found it.
4. **B** - IVH strikes fragile new brain vessels in the first days and often causes no outward signs. Grades 1-2 usually heal well; grades 3-4 are serious. Honest looking lets the team and family know the truth and plan.
5. **B** - In the womb, the placenta is the oxygen supplier and the fluid-filled lungs are under construction — so the ductus arteriosus detours blood right past them. It's supposed to seal within days after birth.
6. **A** - Patent = open. In preemies the detour often fails to close on schedule. The echo measures it; treatment is patience, a vessel-tightening medicine like ibuprofen, or a tiny plug threaded in through a leg vein.
7. **B** - Ibuprofen (and its relatives indomethacin and acetaminophen) block the signals keeping the ductus relaxed, nudging it to tighten shut — in a carefully calculated NICU dose.
8. **B** - X-rays cost a small dose of radiation, so they're used purposefully. Ultrasound is radiation-free and lives at the bedside — the NICU's favorite camera, safe to use every single day.

Final Challenge: Your NICU Rounds (pass mark 9/12)

1. **B** - Preemies are missing every warmth tool at once, and their high skin-to-body ratio leaks heat fast. Isolette with servo control, please — and kangaroo care with mom this afternoon.
2. **C** - For a preemie on oxygen, the target is roughly 90-95%. Parked at 100%, you can't tell how much extra oxygen the blood is soaking in — and too much damages developing eyes (ROP) and lungs. Wean it down.
3. **B** - Textbook A's and B's. Gentle stimulation restarts breathing; daily caffeine keeps the breathing center alert; and the spells fade as the brain matures toward the due date.

4. **A** - Treat babies, not numbers. Some babies with lowish pressure are delivering blood just fine, and overtreating has harms. If the baby shows poor delivery, then fluids, squeeze-medicines, and cause-hunting.
5. **B** - Grunting, flaring, retractions, hazy X-ray: sticky surfactant-poor lungs collapsing with every breath. CPAP keeps the balloons partly inflated; surfactant liquid fixes the stickiness itself.
6. **B** - Feeding is a brain stunt that isn't ready yet, so gavage feeding skips the risky part. Tiny careful feeds also protect the immature gut while it wakes up.
7. **B** - A distended belly and feeds not moving are classic NEC warnings. Feeds stop, the gut rests, X-rays look for gas in the gut wall, antibiotics start – because caught early, outcomes are far better.
8. **A** - Bili lights: blue light bends bilirubin into water-friendly forms (like lumirubin) the body flushes out without the slow newborn liver. Feeding keeps the exit route moving. Recheck to watch the level fall.
9. **B** - Every baby born before about 30-32 weeks gets a screening head ultrasound in week one. The soft spot is the window; silent bleeds are exactly why you look without waiting for symptoms.
10. **D** - Patience, medicine, or a catheter-delivered plug – those are the real tools. The detour vessel was supposed to seal after birth on its own; in preemies it often needs a push.
11. **B** - The Goldilocks lesson: thousands of babies in careful trials taught medicine that even oxygen has a right dose. That's evidence-based medicine – the NICU's core superpower.
12. **B** - Kangaroo care is evidence-backed medicine that parents provide – warmth, stability, milk supply, bonding. The NICU's most advanced technology shares the room with its oldest one: a parent.